

Architecting Aureon: A Next-Generation Hybrid Reconciliation Engine for the Post-Trade Era

1. Executive Summary: The Structural Imperative for Aureon

The global post-trade landscape is undergoing a seismic shift, driven by a confluence of regulatory compression, asset class diversification, and the exponential growth of data volume. The transition to T+1 settlement cycles in North American markets—and the imminent follow-on in the UK and Europe—has effectively removed the operational buffer that financial institutions relied upon for decades. In a T+2 world, a reconciliation break identified on T+1 could be resolved before settlement. In a T+1 environment, that same break represents a failed trade, a regulatory breach, and a direct hit to the balance sheet via CSDR penalties or interest claims.

Simultaneously, the complexity of investment portfolios has outpaced the capabilities of legacy reconciliation platforms. The modern asset manager does not merely trade listed equities; they hold multi-currency OTC derivatives, private credit, and digital assets, often across fragmented liquidity pools in India, APAC, and the West. Legacy systems—characterized by brittle SQL logic, overnight batch processing, and a heavy reliance on manual "ticking and tying" in spreadsheets—are mathematically incapable of handling this velocity and variety. They create "cash drag" by obscuring true liquidity positions and introduce operational risks that no amount of manual headcount can mitigate.

This report presents the architectural blueprint for **Aureon**, an AI-native post-trade operating system designed to solve these structural inefficiencies. The core mandate of Aureon is to deliver a production-grade reconciliation engine that is not merely a passive checker of data but an active, intelligent agent of operational integrity. The architecture is founded on a **hybrid philosophy**: it fuses the deterministic precision of SQL-based rule sets for the "happy path" (80% of volume) with the probabilistic adaptability of Large Language Models (LLMs) and Machine Learning (ML) for exception management (20% of volume, 90% of effort).

The design detailed herein is exhaustive and implementation-ready. It moves beyond high-level concepts to define the specific ingestion mappings for SWIFT MT/MX messages, the exact SQL logic for "one-to-many" cash matching, and the prompt engineering required to resolve complex corporate action discrepancies. It addresses the specific regulatory nuances of key markets—from the Securities Transaction Tax (STT) logic in India to the Wash Sale rules in the US and Stamp Duty in the UK. By partitioning the workload between rigid algorithms and cognitive AI, Aureon offers a pathway to "Continuous Reconciliation,"

transforming the back office from a cost center into a strategic asset that guarantees the financial truth of the enterprise.

2. The Hybrid Architecture: Fusion of Deterministic and Probabilistic Logic

The central thesis of the Aureon architecture is that neither traditional deterministic logic nor modern generative AI is sufficient on its own. Deterministic engines are precise and auditable but brittle; they fail when data formats drift or when unstructured narratives (like bank statement descriptions) obscure the transaction details. Conversely, Generative AI is adaptable and context-aware but non-deterministic and computationally expensive; relying on it for millions of standard matches is inefficient and introduces the risk of hallucination.

Aureon resolves this dichotomy through a layered, "waterfall" architecture. Data flows through a rigorous normalization layer into a high-speed Deterministic Engine. Only the residue—the exceptions—flow into the AI Engine. This ensures that the system is cost-effective, high-speed, and resilient.

2.1 The Data Ingestion and Normalization Layer

The first challenge in any reconciliation system is the "Tower of Babel" problem: every custodian, broker, and bank speaks a slightly different dialect of financial data. Aureon addresses this through an **Extract-Load-Transform (ELT)** pipeline that prioritizes raw data preservation and standardized internal modeling.

2.1.1 Ingestion Framework

Aureon ingests data via three primary channels:

1. **SFTP/File Drop:** For legacy CSV, Excel, and flat files from smaller brokers.
2. **SWIFT Connectivity:** Direct consumption of ISO 15022 (MT) and ISO 20022 (MX) messages.
3. **API/Webhooks:** Real-time JSON streams for modern execution platforms and crypto exchanges.

Raw files are immediately effectively "sealed" in an immutable Data Lake (e.g., Snowflake or S3) to ensure a pristine audit trail. If a transformation rule is later found to be incorrect, the system can "replay" the ingestion from the raw source.¹

2.1.2 The Standard Data Model (SDM)

The crux of normalization is mapping disparate external formats to the Aureon Standard Data Model. This creates a "lingua franca" for the system.

Trade Data Normalization:

Trade data is notoriously fragmented. A "Buy" might be represented as "B," "1," or "BUY" depending on the source.

- *Mapping Logic*: All directional indicators are mapped to a standardized Enum: SIDE_BUY or SIDE_SELL.
- *Identifier Precedence*: The system prioritizes ISIN as the primary key. If ISIN is missing, it falls back to CUSIP, then SEDOL, then Ticker. A "Security Master" lookup service enriches the record: if a broker provides only a Ticker, Aureon fetches the ISIN to ensure matching against the Custodian who uses ISIN.²

SWIFT Message Parsing Strategy:

The engine includes pre-built parsers for standard SWIFT message types, extracting specific tags into the SDM.

SWIFT MT Type	Data Domain	Key Tags Parsed	Aureon SDM Field
MT940	Cash Statement	:61: (Value Date, Amount, D/C mark)	txn_date, amount, direction
		:86: (Information to Account Owner)	narrative_raw (Crucial for AI)
MT950	Cash Statement	:61: (Value Date, Amount)	txn_date, amount
MT535	Holdings/Position	:93B::AGGR (Aggregate Quantity)	qty_total
		:93B::AVAL (Available Quantity)	qty_available
		:93B::PLED (Pledged Quantity)	qty_pledged
MT540-543	Trade Confirm	:35B: (ISIN), :36B: (Qty), :31P: (Price)	security_id, quantity, price

Table 1: SWIFT MT to Aureon SDM Mapping Strategy

The :86: tag in MT940 messages is particularly critical. It often contains unstructured text like "TRF FM AC 1234 REF INV 99". While the deterministic engine might ignore this, Aureon stores

it in narrative_raw for the AI layer to analyze later.³

2.2 Layer 1: The Deterministic Matching Engine

This layer is the workhorse of Aureon. It utilizes set-based SQL logic to match records based on "hard" keys. It is designed to process millions of rows in seconds with zero error rate.

The Pass Strategy:

Matching is not a single query; it is a sequence of "passes," moving from strict to loose criteria.

- **Pass 1 (Perfect Match):** Matches on. This typically clears 60-70% of volume.
- **Pass 2 (Tolerance Match):** Matches on where Amount difference is < \$0.01 (handling FX rounding).
- **Pass 3 (Date Slippage):** Matches on where Trade Date is +/- 1 day (handling timezone differences between APAC and US).³

2.3 Layer 2: The AI/LLM Exception Engine

Records that survive the deterministic passes are "exceptions." These are passed to the AI Engine. This layer uses vector embeddings and LLMs to understand the *semantic* relationship between records.

Mechanism:

1. **Vectorization:** Unmatched records are converted into vector embeddings. The text description "Payment for Invoice 123" is mathematically represented in a way that is "close" to the ledger description "Inv 123 Sett".
2. **Semantic Search:** The AI searches for unmatched items in the opposing dataset that are semantically similar, even if they share no common ID.
3. **Pattern Learning:** If the user manually matches "Global Custodian A" with "Sub-Custodian B" frequently, the ML model learns this association and proposes it automatically in future runs.⁵

3. The Aureon Rule Catalog: Implementation-Ready Logic

The heart of the system is the Rule Catalog. This is not a static list but a library of logic gates that can be composed to handle specific reconciliation types.

3.1 Trade Reconciliation Rules

Trade reconciliation acts as the "front door" control. It ensures that what the Front Office thinks it traded matches what the Street thinks it executed.

3.1.1 Core Matching Keys & Tolerances

The rule engine enforces the following logic for standard equity and bond trades.

Field	Match Logic	Standard Tolerance	Reasoning/Implementation Note
ISIN/CUSIP	Exact	None	Primary identifier. Must link to Security Master to handle Ticker vs. ISIN mappings.
Direction	Exact	None	Buy/Sell. Critical for position keeping. Map '1'/2' to 'BUY'/'SELL'.
Quantity	Exact	None	Quantity breaks are settlement stoppers. Partial fills must be aggregated (see 3.1.2).
Price	Tolerance	+/- 0.01 (Price Ccy)	Accommodates truncation (4 decimals vs 2 decimals).
Net Amount	Tolerance	+/- 1.00 (Settl Ccy)	Allows for minor fee/tax calculation differences.
Trade Date	Window	+/- 1 Day	Accommodates global timezone shifts (e.g., trade in Tokyo booked in NY).
Settlement Date	Exact	None	Critical for cash forecasting. Mismatch triggers

			"Settlement Risk" alert.
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Table 2: Trade Reconciliation Rule Set

3.1.2 Handling Partial Fills (The "Roll-Up" Logic)

A common source of breaks is "Block vs. Allocation." An asset manager executes a block order of 10,000 shares. The broker fills this in three chunks: 2,000, 3,000, and 5,000.

- *Deterministic Rule:* The engine must support "Many-to-One" matching on the Trade side.
- *Pseudocode Logic:*

```
SQL
-- Aggregate Broker Fills by Order ID
SELECT Order_ID, SUM(Quantity) as Total_Qty,
       SUM(Net_Amount) as Total_Amt,
       SUM(Quantity * Price) / SUM(Quantity) as Avg_Price
FROM Broker_File
GROUP BY Order_ID
-- Match against Internal OMS Block
JOIN OMS_File
ON Broker_File.Order_ID = OMS_File.Order_ID
WHERE ABS(Broker_Total_Qty - OMS_Qty) = 0
```

If Order_ID is not available in the broker file (common in street-side FIX drops), the engine groups by and attempts to find a subset of broker trades that sums to the OMS quantity (Subset Sum Problem).⁷

3.1.3 Commission and Fee Logic

Fees are often a source of "noise" breaks.

- *Rule:* Segregate "Gross Price" matching from "Net Amount" matching.
- *Implementation:* Create a separate "Fee Variance" bucket. If Price and Quantity match, but Net Amount differs by < 0.5%, auto-classify as "Fee Break" and route to low-priority queue. This prevents settlement-critical breaks from being obscured by minor P&L discrepancies.²

3.2 Cash Reconciliation Rules

Cash reconciliation is arguably the most complex due to the variety of transaction types (wires, checks, sweeps, fees) and the lack of standardized identifiers.

3.2.1 Matching Algorithms

- **One-to-One:** Matches strictly on Reference_ID (e.g., FedWire IMAD/OMAD numbers) and Amount.

- **One-to-Many (The "Net Settlement" Problem):** A Clearing House (CCP) often sends a single net wire for the day's activity, which corresponds to 50 individual trades in the internal ledger.
 - *Logic:* The engine identifies the "Anchor" transaction (the large bank wire). It then queries the ledger for all unsettled trades with the same Settlement_Date and CCP_ID. It sums these ledger items. If $\text{Sum}(\text{Ledger}) == \text{Bank_Wire}$, it links them as a match group.
- **Many-to-Many:** Complex netting. Aureon uses a heuristic algorithm here. It groups unmatched items by Date and Counterparty, sums them, and checks if the net difference is zero (or within tolerance). If the logic gets too complex (combinatorial explosion), it hands off to the AI layer.⁹

3.2.2 Date Tolerance Logic

Cash timing is fluid. A check issued on Monday (Ledger Date) may clear on Thursday (Bank Date).

- *Rule:* Use a "Sliding Window" match.
- *Logic:* For a Ledger entry on Date T, look for a matching Bank entry between ``.
- *Constraint:* Ensure unique matching. If multiple matching amounts exist within the window, use the one with the closest date or matching narrative.³

3.3 Position Reconciliation Rules

Position reconciliation acts as the ultimate "sanity check" on the portfolio. It validates the cumulative result of all trading and corporate action activity.

3.3.1 Bucket Alignment

Custodians report holdings in specific "buckets." Aureon must map these to the internal IBOR view.

- **Settled Quantity:** Shares actually in the vault.
- **Pending Delivery/Receipt:** Shares traded but not yet settled.
- **Pledged/Segregated:** Shares posted as collateral.
- *Rule:* $\text{IBOR_Total_Position} = \text{Custodian_Settled} + \text{Custodian_Pending_Receipt} - \text{Custodian_Pending_Delivery}$.¹¹

3.3.2 Cost Basis Validation

For tax reporting and P&L, Aureon reconciles the "Unit Cost" or "Total Cost Basis."

- *Logic:* This is critical for catching "Wash Sale" adjustments or incorrect capitalization of fees. A break here doesn't affect settlement but distorts NAV and Tax Liability. The engine flags cost breaks > 0.1% for review by the Tax Operations team.¹³

3.4 NAV Reconciliation (Shadow Accounting)

For Fund Administrators and Hedge Funds, reconciling the Net Asset Value (NAV) is the final,

critical step. Aureon acts as a "Shadow NAV" system.¹⁴

3.4.1 The NAV Control Framework

The engine performs a "Top-Down" and "Bottom-Up" check.

1. **Bottom-Up:** Reconcile every line item (Position MV, Cash, Accruals).
2. **Top-Down:** Compare the final NAV per Share.

3.4.2 Stale Pricing Detection

A common issue in NAV accuracy is "Stale Pricing," where an illiquid asset's price isn't updated, or a data feed fails.

- *Rule:* If $\text{Current_Price} == \text{Previous_Day_Price}$ AND $\text{Asset_Class} \neq \text{'Private_Equity'}$, trigger warning.
- *Logic:* For liquid equities, unchanged prices for 2+ days are highly suspicious. The engine checks market volume; if $\text{Volume} > 0$ but Price is unchanged, it raises a "Stale Price" alert.¹⁵

3.4.3 Large Movement Logic

- *Rule:* Trigger alert if $(\text{Current_NAV} - \text{Prev_NAV}) / \text{Prev_NAV} > \text{Threshold}$ (e.g., 5% for equity, 1% for fixed income).
- *Context:* This catches "Fat Finger" errors or significant market dislocations that require explanation in the investor report.¹⁷

4. Regional Specifics: US, UK, India, and APAC

Aureon's global capability is defined by its library of jurisdiction-specific rules.

4.1 United States: The T+1 Pressure Cooker

The US market's move to T+1 settlement¹⁸ imposes strict timelines.

- **Affirmation Rule:** Institutional trades must be affirmed by 9:00 PM ET on Trade Date (T+0).
 - *Aureon Logic:* The engine runs a "T+0 Control" sweep at 8:00 PM ET. Any trade in the OMS that lacks a matching confirmation from the DTCC/Idom system is flagged as "Critical - At Risk of Fail." This allows ops teams 1 hour to remediate.¹⁹
- **SEC Section 31 Fees:** Sellers pay a transaction fee to the SEC.
 - *Calculation:* $\text{Trade_Value} * \text{SEC_Fee_Rate}$ (e.g., \$8 per million, updated annually).
 - *Rule:* The engine calculates this expected fee and reconciles it against the "Regulatory Fee" column in the broker confirm. A mismatch implies an incorrect fee rate was applied by the broker.²⁰
- **Wash Sales:** The IRS disallows losses on securities sold and repurchased within 30 days.
 - *Rule:* The engine tracks realized losses. If a matching CUSIP is bought within 30 days, it flags the P&L entry as "Disallowed Loss" and adjusts the Cost Basis of the new lot

upwards.¹³

4.2 India: The Complexity of Statutory Levies

Indian markets (NSE/BSE) are unique due to the multifaceted tax structure involved in trading.

- **Securities Transaction Tax (STT):**
 - *Logic:* STT rates vary by trade type.
 - Delivery Equity: 0.1% on Buy & Sell.
 - Intraday Equity: 0.025% on Sell only.
 - Futures: 0.02% on Sell.
 - Options: 0.1% on Premium (Sell).²²
 - *Reconciliation:* Aureon classifies the trade (Intraday vs Delivery) based on the OMS "Product Type" tag and recalculates the STT. It matches this against the broker's charge. Mismatches often indicate incorrect trade tagging (e.g., an intraday trade booked as delivery).
- **GST & Stamp Duty:**
 - *Rule:* GST (18%) applies to the *Brokerage* amount, not the trade value. Stamp duty is state-specific but unified rates apply for exchange trades. Aureon validates these line items separately to prevent tax leakage.²⁴

4.3 United Kingdom: SDRT and CREST

- **Stamp Duty Reserve Tax (SDRT):**
 - *Rule:* 0.5% tax on purchase of UK equities.
 - *Logic:* Aureon calculates $0.005 * \text{Trade_Value}$ for all Buy trades with `Country_of_Issue = GB`. It reconciles this against the CREST settlement instruction.
- **CREST Penalties:**
 - *Rule:* Since CSDR implementation, settlement fails incur daily cash penalties. Aureon ingests "Penalty Credit/Debit" lines from the custodian and reconciles them against internal "Failed Trade" logs to verify the penalty amount.²⁶

4.4 APAC: Fragmentation and FX Controls

- **FX Control Rules:** Markets like Korea, Taiwan, and Indonesia have strict currency controls.
 - *Rule:* Trades must be linked to a specific FX transaction (Pre-funding). Aureon links the Security Trade ID to the FX Trade ID. A break in this link triggers a "Funding Risk" alert, as the trade may fail due to lack of local currency.
- **Settlement Cycles:** While India is T+1 (and moving to T+0), other APAC markets may be T+2. The engine maintains a "Market Settlement Calendar" to apply the correct "Expected Settlement Date" logic for each market.

5. Advanced Domain Logic: Corporate Actions and FX

5.1 Corporate Actions: The Lifecycle of a Break

Corporate actions (dividends, splits, mergers) are the single largest source of breaks because they alter the fundamental attributes of the position (Price, Qty, or Cash).

5.1.1 Dividend Market Claims (T+1 Impact)

In a T+1 or T+2 world, "Market Claims" arise when a trade executes "Cum-Dividend" but settles "Ex-Dividend."

- **Scenario:** Fund A buys 1000 Apple shares on Cum-Date. The trade fails to settle until after Record Date. The Seller receives the dividend, but Fund A is entitled to it.
- **Aureon Logic:**
 1. **Detection:** Identify trades where $\text{Trade_Date} < \text{Ex_Date}$ AND $\text{Actual_Settlement_Date} \geq \text{Record_Date}$.
 2. **Claim Generation:** The engine automatically generates a "Pending Market Claim" receivable in the cash ledger: $\text{Qty} * \text{Dividend_Rate}$.
 3. **Matching:** This generated claim is then matched against the eventual "Compensation Payment" received from the counterparty/broker. This automation prevents manual "claim chasing".²⁷

5.1.2 Stock Splits

- **Rule:** On Ex-Date of a 2-for-1 split, Aureon applies a transformation rule: $\text{New_Qty} = \text{Old_Qty} * 2$, $\text{New_Price} = \text{Old_Price} / 2$.
- **Validation:** It compares the $\text{Total_Market_Value}$ (which should remain roughly constant) to ensure the split was applied correctly and not treated as a massive price drop.¹³

5.2 FX and CLS Settlement

For cross-border investing, FX settlement risk is paramount.

- **CLS Matching:** Continuous Linked Settlement (CLS) minimizes risk but creates reconciliation complexity (netting).
- **Logic:** Aureon aggregates all FX trades settling in CLS for a given currency pair and value date. It compares the Net_Pay_In or Net_Pay_Out against the CLS bank statement.
- **Tolerance:** High tolerance for "Pips" (small price movements) is required due to the difference between the "booked rate" and the "average rate" often applied in CLS netting.³

6. AI Partitioning: Self-Healing and Exception Management

While SQL handles the rules above, Aureon's AI layer handles the messiness of reality.

6.1 Unstructured Data Ingestion (LLM-based)

Often, a broker sends a PDF contract note or an email about a private placement.

- *Workflow:*
 1. **OCR:** Extracts raw text from the document.
 2. **LLM Prompt:** "Extract the Trade Date, ISIN, Quantity, Price, and Settlement Amount from the following text. Output as JSON conforming to schema {date: YYYY-MM-DD,...}."
 3. **Validation:** The output is fed into the SDM. If the LLM's confidence score is low, it flags for human review.²⁸

6.2 Self-Healing Pipelines

Data formats change. A broker might rename a CSV column from ISIN to Security_Ref without notice. In a legacy system, this breaks the nightly batch.

- *Aureon Mechanism:* The ingestion engine detects the schema error. It passes the new header row to an LLM.
- *Prompt:* "The expected column 'ISIN' is missing. The available columns are 'TradeID, Security_Ref, Price'. Which column is most likely the ISIN?"
- *Action:* The LLM identifies Security_Ref as the likely candidate. The system dynamically updates the mapping, retries the load, and logs a warning for the engineer. The process heals itself without downtime.²⁹

6.3 Semantic Exception Resolution

For unmatched cash items, the AI analyzes the narrative.

- *Scenario:* Ledger has "Payment for Inv 999". Bank has "ACH TFR REF: 999 - Fees".
- *AI Logic:* The AI calculates the vector distance between descriptions. It notices the common "999". It checks if the amount difference equals a standard fee (e.g., \$15). It proposes a match with the explanation: "Match found with \$15 fee deduction."⁶

7. Implementation Roadmap

Deploying Aureon is a journey from deterministic stability to AI-driven autonomy.

Phase 1: Foundation & Deterministic Core (Months 1-3)

- **Objective:** Establish ingestion and SQL matching.
- **Actions:**
 - Build ELT pipelines for SWIFT MT940, MT535, and standard Broker CSVs.
 - Implement standard 1-to-1 matching for Cash and Trades.
 - Configure US and UK regional rules (T+1 alerts, SDRT).
- **Success Metric:** 75% Auto-match rate.

Phase 2: Advanced Logic & Global Expansion (Months 4-6)

- **Objective:** Handle complex markets and events.
- **Actions:**
 - Deploy Corporate Action lifecycle logic (Market Claims).
 - Implement India STT/GST logic and APAC FX controls.
 - Roll out NAV "Shadow Accounting" validation (Stale Price checks).
- **Success Metric:** 90% Auto-match rate; Support for all asset classes.

Phase 3: AI Augmentation & Continuous Recs (Months 7-12)

- **Objective:** Automate the "Last Mile" of exceptions.
- **Actions:**
 - Train LLM on historical break data to learn firm-specific patterns.
 - Deploy Self-Healing pipelines for schema drift.
 - Activate "Generative Explanations" for breaks to aid human operators.
- **Success Metric:** 98% Auto-match rate; 50% reduction in manual break investigation time.

8. Conclusion

Aureon represents the future of the post-trade function. By acknowledging that reconciliation is a "two-speed" problem—requiring high-speed deterministic logic for the bulk and high-intelligence AI for the exceptions—it solves the scalability crisis facing modern finance. In a T+1 world, Aureon provides the requisite speed; in a complex global market, it provides the necessary intelligence. It transforms reconciliation from a retrospective administrative burden into a real-time, strategic control framework.

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